Go for JavaScript Developers

From React and Next.js to Go, and building real applications with Grit

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Written against Grit v3.403.0 and Go 1.25

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Why Go, when you already ship JavaScript

You already build and ship software. You know React, you know Next.js, and you can take an idea to production without asking anybody how. So the first question this book owes you is not "what is Go", it is "why would you spend a month learning a second language when the one you have works".

Here is the honest answer, and then the honest list of what you give up.

What actually pushed you here

Most JavaScript developers do not come to Go out of curiosity. They come because something in their stack got painful in a specific way. Usually one of these four.

The deployment got heavy. Your Next.js app needs Node on the server, a `node_modules` directory with a few hundred megabytes in it, and a build step that has to run somewhere. You want to copy one file onto a server and run it. A compiled Go program is one file. It has no runtime to install, no package directory, and no build step on the target machine. You build it on your laptop and the output is the thing that runs.

The work stopped being request-and-respond. An API that answers a question in forty milliseconds is a perfect job for Node. A process that holds ten thousand open connections, or runs a job queue, or chews through a large file while staying responsive, is where the single-threaded event loop stops being an elegance and starts being a constraint you work around.

You wanted the errors earlier. In JavaScript, a renamed field is found by a user. TypeScript helps enormously and you should keep using it, but it is a layer on top of a language that does not have types, and it disappears at runtime. In Go the types are the language. A program that compiles has already had a whole class of mistake removed from it.

Somebody said the word "backend" and meant it. Payment handling, a scheduler, an audit trail, something that must still be correct at 3am when nobody is watching. Go was built by people who operate systems like that for a living, and it shows in what the language refuses to let you do.

> **IF YOU KNOW JAVASCRIPT**
> None of this makes Go better than JavaScript. It makes it different in ways that matter for one kind of work. You are not replacing React. In the second half of this book you will have a Go backend and a React frontend in the same project, talking to each other with types generated from one definition. That is the shape most people actually want.

The shape of Go, in one page

Before any syntax, here is the character of the language, because it explains every decision you are about to meet.

Go is **small on purpose**. The entire specification is about one hundred pages. There are twenty five keywords. JavaScript has roughly double that and a standard library that has grown for thirty years. You can hold all of Go in your head, which is a design goal rather than an accident, and it is why Go teams onboard people in days rather than weeks.

Go is **explicit on purpose**. There is no implicit type coercion, no `this` that changes meaning depending on how a function was called, no prototype chain to walk. When you read a line of Go, what happens is what is written. This is sometimes tedious and it is never surprising, and after a few weeks you will notice you have stopped debugging a category of problem entirely.

Go is **opinionated about formatting**, which sounds trivial and is not. There is one correct way to format Go code, a tool called `gofmt` produces it, and every Go project on earth uses it. No Prettier config, no debate about semicolons, no `.editorconfig`. The argument is over because the language ended it.

Go **compiles fast**. Not “fast for a compiled language”, fast in absolute terms: a medium project builds in a couple of seconds. This matters more than it sounds, because it means the compile step never becomes a reason to avoid checking your work.

What you give up, stated plainly

A book that only lists advantages is marketing. Here is the other side, and you should weigh it before chapter two.

You will write more lines. Go has no `map`, `filter` or `reduce` built into the language the way JavaScript does. Where you write one chained expression, you will often write a four-line loop. Go’s position is that the loop is easier to read six months later. You may disagree, and you will still write the loop.

Error handling is verbose. You are about to meet `if err != nil` and you will type it hundreds of times. There is no `try` / `catch` wrapping half a function. Every single

operation that can fail says so, at the point it can fail. This is the single biggest adjustment coming from JavaScript, and chapter four is entirely about making peace with it.

The ecosystem is smaller. npm has over two million packages. Go has far fewer. For web application work this matters less than you would think, because Go's standard library covers an enormous amount of what you would reach for a package to do in Node. But if you need an obscure SDK, check before you commit.

No null safety. Go has `nil`, and a nil pointer dereference will crash your program at runtime exactly like reading a property of `undefined` crashes yours. Go's type system is simpler than TypeScript's, and in this one specific area TypeScript is genuinely ahead.

> **WATCH OUT**
> If someone tells you Go is simply better than JavaScript, stop listening to them. The two languages were built for different jobs by people with different problems. Go's concurrency is better than the event loop for sustained work, and the event loop is a beautiful fit for a UI. Use both.

The one example that explains everything

Here is the same program in both languages: read three URLs, and print how long each took. It is worth studying before you understand any Go syntax, because the difference in shape is the whole argument.

JAVASCRIPT

```
async function timeAll(urls) {
  const results = await Promise.all(
    urls.map(async (url) => {
      const start = Date.now();
      await fetch(url);
      return { url, ms: Date.now() - start };
    })
  );
  for (const r of results) {
    console.log(r.url, r.ms);
  }
}
```

GO

```
func timeAll(urls []string) {
    var wg sync.WaitGroup
    results := make([]result, len(urls))

    for i, url := range urls {
```

```
        wg.Add(1)
        go func() {
            defer wg.Done()
            start := time.Now()
            http.Get(url)
            results[i] = result{url, time.Since(start)}
        }()
    }

    wg.Wait()
    for _, r := range results {
        fmt.Println(r.url, r.ms)
    }
}
```

The JavaScript is shorter and, if you know JavaScript, clearer. Do not let anybody tell you otherwise.

But look at what the Go version is doing. The word `go` in front of that function call starts a **goroutine**: an independent line of execution that the Go runtime schedules onto a real operating system thread. Three URLs here, but the same code with fifty thousand URLs would start fifty thousand goroutines, and Go would handle it, because a goroutine costs about two kilobytes of memory rather than the megabyte an OS thread costs.

The JavaScript version is also concurrent, and for three HTTP requests it is just as fast, because all three are waiting on the network and the event loop is excellent at waiting. The difference appears when the work is not waiting. If each of those tasks had to parse a large document or resize an image, the JavaScript version would do them one at a time on one core, and the Go version would use every core you have.

> **THE MENTAL MODEL SHIFT**
> In JavaScript, concurrency is something you **await**, and parallelism is something you get by starting more processes. In Go, concurrency is something you **start**, with one keyword, and parallelism comes free because the runtime spreads goroutines across cores. Chapter eleven is where this clicks properly. For now, notice only that `go doSomething()` is the entire syntax.

What this book does

This book has two halves and they have different jobs.

Part one through three teach you Go, assuming you know JavaScript and nothing else. Every concept is introduced by naming the JavaScript thing it resembles, then

showing exactly where the resemblance breaks. You will build small things constantly: a CLI tool, a JSON API with no framework at all, a URL shortener backed by a real database. By the end you can read and write Go without this book open.

Part four and five teach you Grit, which is a framework that generates a Go API, a React admin panel and the typed client between them from one description of your data. The reason it comes second is that a framework you cannot read is a liability. Once you know Go, Grit's generated code is just Go you did not have to type, and you can change any of it.

> **WHAT YOU WILL HAVE BUILT BY THE END**
> A command line tool that formats JSON. An HTTP API written with nothing but the standard library. A URL shortener with a SQLite database. A task tracker, a blog with a working admin panel, and an invoicing application with customers, line items and money handled correctly. All of them small, all of them finished, none of them a toy that stops at the interesting part.

How to read it

Type the code. Do not copy and paste it. This is advice everybody gives and almost nobody follows, so here is the specific reason it matters for Go: the compiler's error messages are a large part of how you learn the language, and you only see them if you make mistakes. Typing produces mistakes. Pasting does not.

Run everything. Every code block that is a complete program is marked, and you should run it, break it deliberately, and run it again. Go builds in seconds, so the cost of an experiment is almost nothing.

Do the "Your turn" boxes. They are short, and they are the difference between recognising Go and writing it.

> **YOUR TURN**
> Before the next chapter, answer this for your own situation in two sentences: what is the specific thing in your current stack that made you pick this book up? Write it down. In chapter twenty three there is a list of twenty six problems that every project runs into, and you will want to know which of them is yours.

0

Installing Go and your first program

This chapter gets Go on your machine, gets a program running, and introduces the toolchain. If you have ever set up a Node project you will find this shorter than you expect, because most of the steps you are used to do not exist.

Installing

Go to go.dev/dl and download the installer for your platform. There is no version manager to install first, no equivalent of `nvm`, and for most people there never needs to be. Go's compatibility promise means a program written for Go 1.0 in 2012 still compiles today, so "which version" is rarely a question that costs you an afternoon.

Check it worked:

```
go version
```

You should see something like `go version go1.25.0 windows/amd64`. If the command is not found, the installer did not put Go on your `PATH`; restart your terminal first, because that fixes it more often than anything else.

> **IF YOU KNOW JAVASCRIPT**
> There is no `nvm`, no `.nvmrc`, and no per-project runtime. One Go installation builds every project on your machine. When a project needs a specific version, it says so in one line of `go.mod` and the toolchain downloads that compiler itself.

Your first program

Make a directory and put one file in it.

```
// main.go
package main

import "fmt"

func main() {
```

```
    fmt.Println("Hello from Go")
}
```

Run it:

```
go run main.go
```

Four things in that tiny file deserve explanation, because each is a rule you will meet constantly.

`package main` declares which package this file belongs to. Every Go file starts with a package declaration. The name `main` is special: it means this package builds into a runnable program rather than a library.

`import "fmt"` pulls in the formatting package from the standard library. Note there is no `npm install`. `fmt` ships with Go, as does an HTTP server, a JSON encoder, a test runner, a crypto suite and a templating engine.

`func main()` is the entry point. A program starts here. There is exactly one `main` function in a program.

`fmt.Println` prints a line. Capital `P` matters enormously, and it is the next section.

Capital letters are the access modifier

This is the first genuinely surprising rule in Go, and it catches everybody.

An identifier starting with a capital letter is exported. One starting with a lowercase letter is private to its package. That is the entire visibility system. There is no `public`, no `private`, no `export` keyword.

```
package money

// Convert is callable from other packages, because it is capitalised.
func Convert(cents int) string { ... }

// round is not. Only code inside package money can call it.
func round(f float64) int { ... }
```

> **IF YOU KNOW JAVASCRIPT**
> In JavaScript you choose what leaves a module with `export`. In Go the name itself decides. `fmt.Println` works because `Println` is capitalised inside package `fmt`; if it were `println` you could not reach it. This applies to functions, types, struct fields, constants and variables alike.

> **WATCH OUT**
> This bites hardest with struct fields and JSON. A lowercase field is invisible to the JSON encoder, because the encoder lives in another package and genuinely cannot see it. A struct whose fields are all lowercase encodes to `{}`, with no error and no warning. Chapter sixteen covers this properly, but remember the symptom now: empty JSON usually means lowercase fields.

Modules: go.mod is your package.json

A single file is fine for a demo. A real project is a **module**. Initialise one:

```
mkdir hello && cd hello
go mod init github.com/yourname/hello
```

That writes `go.mod`:

```
module github.com/yourname/hello

go 1.25
```

That is the whole file. Compare it to a `package.json` for an equivalent project and notice what is missing: no `scripts`, no `devDependencies`, no `main`, no `type: module`. The module path doubles as the import path, which is why it looks like a URL: if you push this to GitHub, anybody can import it at exactly that address with no registry publish step.

Add a dependency and `go.mod` grows a line:

```
go get github.com/google/uuid
```

```
module github.com/yourname/hello

go 1.25

require github.com/google/uuid v1.6.0
```

> **IF YOU KNOW JAVASCRIPT**
> `go.sum` is your lockfile, equivalent to `package-lock.json`. Commit it. There is no `node_modules`: downloaded packages go to a shared module cache outside your project, and the compiler links what it needs directly into the binary. Deleting a project directory does not take 300MB of dependencies with it, because they were never there.

The commands you will actually use

Six commands cover almost everything.

`go run .`	Compile and run the current package, without keeping the binary
`go build`	Compile to an executable in the current directory
`go test ./...`	Run every test in this module
`go mod tidy`	Add what is imported, remove what is not. Run it often
`gofmt -w .`	Format every file. Your editor probably does this on save
`go vet ./...`	Report suspicious code the compiler allows

The `./...` pattern means "this directory and everything under it", and you will type it constantly.

Compiling a real binary

This is the part that sells Go to people who have deployed Node.

```
go build -o hello .
./hello
```

You now have one executable file. It contains your code, every dependency, and the Go runtime. The target machine needs nothing installed. You can also build for a platform you are not on:

```
GOOS=linux GOARCH=amd64 go build -o hello-linux .
```

That produces a Linux binary from a Mac or Windows laptop, with no container, no cross-compilation toolchain and no configuration. Copy it to a Linux server and it runs.

PROJECT 1: A GREETING WITH ARGUMENTS

Write a program that takes a name as a command line argument and greets it, falling back to "world" when none is given. You need `os.Args`, which is a slice of strings where `os.Args[0]` is the program name itself.

```
package main

import (
    "fmt"
    "os"
)
```

```
func main() {
    name := "world"
    if len(os.Args) > 1 {
        name = os.Args[1]
    }
    fmt.Printf("Hello, %s\n", name)
}
```

Build it, run `./hello Ada`, then run it with no argument. Then break it deliberately: change `len(os.Args) > 1` to `len(os.Args) > 0` and run it with no argument to see what a real Go panic looks like. Reading panics early is worth more than avoiding them.

YOUR TURN

Extend the greeting to accept several names and greet each on its own line. You will need a `for` loop over a slice, which looks like `for _, n := range os.Args[1:] { ... }`. The next chapter explains every piece of that line, so guessing now is fine and the compiler will tell you when you are wrong.

0

Variables, types, and the compiler that argues with you

This is the chapter where Go feels most foreign, and it is worth knowing why in advance. In JavaScript, a variable is a box that holds anything and can hold something else tomorrow. In Go, a variable has one type forever, decided when it is created, and the compiler refuses to build your program if you break that promise.

That refusal is the feature. Work with it for a week and you will notice a whole category of bug has stopped happening to you.

Declaring things

Go has two ways to make a variable, and you will use one of them ninety percent of the time.

```
var name string = "Ada"   // the long form, type stated
var age = 36              // type inferred, still a var
name := "Ada"             // short form: declare and infer, inside
functions
```

The `:=` operator is the one you want. It declares a new variable and infers its type from the value on the right. It only works inside a function, which is why package level variables use `var`.

> **IF YOU KNOW JAVASCRIPT**
> `:=` is closest to `let` in that it creates a new binding. There is no equivalent of `const` for variables created with `:=`. Go's `const` exists, but it is only for values known at compile time: numbers, strings, booleans. You cannot declare a constant struct or slice. If you want "do not reassign this", Go's answer is to write a small enough function that it is obvious.

An unused variable is a **compile error**, not a warning.

```
func main() {
    count := 10
```

```
    fmt.Println("hello")
}
// ./main.go:4:5: declared and not used: count
```

This feels hostile on day one and reasonable by day ten. Dead variables are usually the residue of a half-finished edit, and Go would rather stop you than let the residue accumulate. The same rule applies to unused imports.

> **WATCH OUT**
> The escape hatch is `_`, the blank identifier, which means "I know there is a value here and I am deliberately ignoring it". You will use it constantly in loops and multiple returns. Do not reach for it to silence a real warning: `_ = count` compiles and tells the next reader nothing.

The types you will actually use

Go has a lot of numeric types. You will use about four of them.

Type	JavaScript	Notes
`string`	`string`	Immutable, UTF-8 bytes. Indexing gives a byte, not a character
`int`	`number`	Whole numbers. 64 bit on any machine you own
`float64`	`number`	Decimals. The default for a fractional literal
`bool`	`boolean`	`true` or `false`. No truthiness, ever
`byte`	-	Alias for `uint8`. A byte of data
`rune`	-	Alias for `int32`. One Unicode code point

The one to notice is that JavaScript's single `number` splits into `int` and `float64`, and **Go will not convert between them for you**.

```
count := 10          // int
price := 2.5         // float64
total := count * price
// invalid operation: count * price (mismatched types int and float64)
```

You must say what you mean:

```
total := float64(count) * price   // 25
```

> **IF YOU KNOW JAVASCRIPT**
> JavaScript coerces almost anything: `"3" * 2` is `6`, `[] + {}` is a string, and `0.1 + 0.2` is famously not `0.3`. Go does none of this. There is no implicit conversion between any two types, not even `int` and `int64`. Every conversion is written out, which is more typing and removes an entire family of surprise.

Zero values: there is no undefined

Every type in Go has a **zero value**, and a declared variable always has it. There is no `undefined`, and no uninitialised memory.

`string`	`""`
`int`, `float64`	`0`
`bool`	`false`
pointers, slices, maps, interfaces, functions	`nil`

```
var name string
var count int
var ready bool
fmt.Println(name == "", count == 0, ready == false) // true true true
```

This is genuinely useful. A struct you create is immediately valid, with every field at a sensible default, and you never check whether a field "exists".

> **WATCH OUT**
> It is also the one place zero values hurt: you cannot tell "the caller set this to zero" apart from "the caller did not set it". If that distinction matters, for example a form field where empty and absent mean different things, you need a pointer (`*int`, where `nil` means absent) or a dedicated type. Chapter eight covers pointers properly. This problem is real, and it is the usual reason you see `*string` in a struct.

No truthiness

An `if` in Go takes a `bool`. Not a value that is bool-ish. A `bool`.

JAVASCRIPT

```
if (name) {
  greet(name);
}
if (items.length) {
```

```
  render(items);
}
```

GO

```go
if name != "" {
    greet(name)
}
if len(items) > 0 {
    render(items)
}
```

More characters, and no memorising which values are falsy. There is no `==` and `===` distinction either, because there is only one `==` and it does not coerce.

Strings are bytes, and that matters

A Go string is a read-only slice of bytes holding UTF-8. Indexing it gives you a byte.

```go
s := "héllo"
fmt.Println(len(s))     // 6, not 5: é takes two bytes
fmt.Println(s[1])       // 195, a byte value, not "é"
```

To work with characters, range over the string, which decodes runes:

```go
for i, r := range "héllo" {
    fmt.Printf("%d: %c\n", i, r)
}
// 0: h
// 1: é      <- index jumps to 3 next, because é was two bytes
// 3: l
// 4: l
// 5: o
```

> **IF YOU KNOW JAVASCRIPT**
> JavaScript strings are UTF-16 and have the same class of problem: `"👍".length` is `2`. Both languages make you think about encoding the moment you leave ASCII. Go is at least explicit that you are holding bytes.

Strings are immutable, so building one in a loop with `+=` allocates a new string every time. For anything beyond a few concatenations use a builder:

```go
var b strings.Builder
for _, word := range words {
```

```
    b.WriteString(word)
    b.WriteString(" ")
}
result := b.String()
```

Control flow, briefly

`if` and `for` are almost familiar, with two differences worth stating.

There are no parentheses around the condition, and braces are mandatory even for one line.

```
if count > 10 {
    fmt.Println("plenty")
} else if count > 0 {
    fmt.Println("some")
} else {
    fmt.Println("none")
}
```

`for` is the only loop keyword. It covers every case JavaScript spreads across `for`, `while` and `for...of`.

```
for i := 0; i < 10; i++ { }      // classic
for count < 10 { }               // while
for { }                          // infinite
for i, v := range items { }      // for...of, with the index
```

An `if` can carry a statement before the condition, scoped to the `if`. You will see this constantly in real code:

```
if err := doSomething(); err != nil {
    return err
}
// err does not exist here, which is the point
```

IF YOU KNOW JAVASCRIPT

`switch` in Go needs no `break`: cases do not fall through. You can switch on no value at all, which makes it a tidy replacement for a long `else if` chain:

```
switch {
case score > 90:
    grade = "A"
```

```
case score > 80:
    grade = "B"
default:
    grade = "C"
}
```

PROJECT 2: A TEMPERATURE TABLE

Write a program that prints a table converting Celsius to Fahrenheit from minus 40 to 100 in steps of 20. The formula is `f = c*9/5 + 32`.

```
package main

import "fmt"

func main() {
    fmt.Printf("%8s %8s\n", "C", "F")
    for c := -40; c <= 100; c += 20 {
        f := float64(c)*9/5 + 32
        fmt.Printf("%8d %8.1f\n", c, f)
    }
}
```

Two things to notice. `float64(c)` is required: without it you get integer division and wrong answers for anything not divisible by five, which is the bug this chapter exists to prevent. And `%8.1f` means "right align in eight columns, one decimal place", which is how Go does string formatting.

Now break it on purpose: remove the `float64(...)` conversion and read the error. Then change it to `c*9/5` with `c` still an `int` and see that it compiles but produces wrong numbers. The compiler catches the type error. It cannot catch the maths error, and knowing which is which is the skill.

YOUR TURN

Add a third column showing Kelvin (`k = c + 273.15`), and right align the header over it. Then make the step size a variable at the top of `main` and confirm the table still lines up when you change it to 10.

0

Functions, multiple returns, and the end of try/catch

This is the chapter that decides whether you get on with Go. Error handling is where the language is most unlike JavaScript, it is the thing newcomers complain about loudest, and the complaint is half right. Let us do the mechanics first and the argument second.

Functions

```
func add(a int, b int) int {
    return a + b
}

func add(a, b int) int {   // same thing, shared type
    return a + b
}
```

The type comes **after** the name, which reads backwards for about a day and then stops mattering. The reason is that it makes complicated declarations read left to right instead of spiralling outwards, which anyone who has read a C function pointer declaration will appreciate.

Functions are values, so the things you do in JavaScript still work:

```
double := func(n int) int { return n * 2 }

func apply(nums []int, f func(int) int) []int {
    out := make([]int, len(nums))
    for i, n := range nums {
        out[i] = f(n)
    }
    return out
}

apply([]int{1, 2, 3}, double)  // [2 4 6]
```

> **IF YOU KNOW JAVASCRIPT**
> That `apply` is `Array.prototype.map`, written by hand. Go had no generics until 1.18 and still has no `map`, `filter` or `reduce` in the standard library for slices. You write the loop. Go's argument is that a three-line loop needs no explanation and a chain of five callbacks sometimes does. You are allowed to find this annoying.

Two return values, and why

A Go function can return several values, and this is not a curiosity: it is the foundation of the error system.

```
func divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, errors.New("divide by zero")
    }
    return a / b, nil
}
```

The caller gets both:

```
result, err := divide(10, 2)
if err != nil {
    fmt.Println("failed:", err)
    return
}
fmt.Println(result)
```

The convention is absolute, across the standard library and every package you will use: **the error is the last return value, and you check it immediately**.

> **IF YOU KNOW JAVASCRIPT**
> JavaScript has one return value, so a function that can fail either throws or returns something like `{ data, error }`. Go makes the second shape the language default. If you have written Go-style error handling in TypeScript with a `Result` type, or used Rust, you already have the mental model.

`error` is just an interface

There is nothing magic about errors. `error` is an interface with one method:

```
type error interface {
    Error() string
}
```

Anything with an `Error() string` method is an error. That means you can make your own, carrying whatever fields you want:

```
type ValidationError struct {
    Field   string
    Message string
}

func (e ValidationError) Error() string {
    return e.Field + ": " + e.Message
}

func validate(age int) error {
    if age < 0 {
        return ValidationError{Field: "age", Message: "must not be
negative"}
    }
    return nil
}
```

Wrapping: adding context without losing the original

A bare error from deep in a call stack is useless on its own. `fmt.Errorf` with the `%w` verb wraps one error inside another, keeping both.

```
func loadConfig(path string) (*Config, error) {
    data, err := os.ReadFile(path)
    if err != nil {
        return nil, fmt.Errorf("loading config from %s: %w", path, err)
    }
    ...
}
```

The message becomes a trail that reads from the outside in:

```
loading config from /etc/app.json: open /etc/app.json: no such file or
directory
```

Use `%w`, not `%v`. `%v` formats the error as text and throws the original away, and then both of the checks below stop working.

Two functions read a wrapped chain. `errors.Is` asks "is this error, anywhere in the chain, that specific one":

```
if errors.Is(err, os.ErrNotExist) {
    // the file was missing, however deeply that was wrapped
}
```

`errors.As` asks "is there an error of this type in the chain", and gives it to you with its fields intact:

```
var ve ValidationError
if errors.As(err, &ve) {
    fmt.Println("bad field:", ve.Field)
}
```

WATCH OUT

`err != nil` is the only check that matters, and comparing an error to `nil` has one notorious trap. A function returning a **concrete** error type rather than the `error` interface can produce an error value that is not `nil` even when it holds nothing:

```
func bad() error {
    var e *ValidationError   // nil pointer
    return e                 // not nil! it is a non-nil interface
holding a nil pointer
}
```

The fix is a rule: **always declare the return type as `error`, and return a literal `nil`, never a typed nil variable.** This catches experienced people and it is worth reading twice.

The argument about verbosity

Here is the complaint, stated fairly. This:

JAVASCRIPT

```
async function setup() {
  try {
    const cfg = await loadConfig();
    const db = await connect(cfg.dsn);
    await migrate(db);
    return db;
  } catch (e) {
    throw new Error(`setup failed: ${e.message}`);
  }
}
```

GO

```
func setup() (*DB, error) {
    cfg, err := loadConfig()
    if err != nil {
        return nil, fmt.Errorf("setup: %w", err)
    }
    db, err := connect(cfg.DSN)
    if err != nil {
        return nil, fmt.Errorf("setup: %w", err)
    }
    if err := migrate(db); err != nil {
        return nil, fmt.Errorf("setup: %w", err)
    }
    return db, nil
}
```

The Go is twice as long. That is not a matter of opinion.

What you get for it is that every failure point is visible at the point it can fail. In the JavaScript, which of the three awaited calls threw? You cannot tell from the code, and the catch block treats all three identically whether or not that is what you wanted. In the Go, each failure is a separate decision, and you could handle the migration failure differently from the connection failure by changing one block.

The honest summary: `try` / `catch` is better when you want to treat a whole region of code as one unit that either works or does not. Explicit returns are better when failures deserve individual handling. Most real backend code is the second kind more often than people expect, and Go bet the language on it.

> **WHAT YOU LOSE, AND SHOULD KNOW ABOUT**
>
> In JavaScript an uncaught exception unwinds to the nearest handler, so a forgotten `catch` still surfaces somewhere. In Go a forgotten error check silently continues with a zero value. Nothing warns you.
>
> This is a real hazard and the tooling answers it: `errcheck`, bundled into `golangci-lint`, fails the build on an unchecked error. Install it early. It is the one linter that is not optional.

Named returns and `defer`

A function can name its return values, which is occasionally useful and frequently overused:

```
func split(sum int) (x, y int) {
    x = sum * 4 / 9
```

```
    y = sum - x
    return          // a "naked" return, sending back x and y
}
```

Use naked returns only in short functions. In a long one, a bare `return` forces the reader to scroll up to learn what is being returned.

Named returns do earn their place with `defer`, which schedules a call to run when the surrounding function exits, no matter how it exits:

```
func process(path string) (err error) {
    f, err := os.Open(path)
    if err != nil {
        return err
    }
    defer f.Close()     // runs on every return path below

    // ... many returns, all of which close the file
    return nil
}
```

`defer f.Close()` immediately after a successful open is one of the most common patterns in Go, and it is how the language avoids the cleanup bugs that `try` / `finally` exists to solve.

WATCH OUT

Deferred calls run in last-in, first-out order, and their **arguments** are evaluated immediately, not at exit:

```
for i := 0; i < 3; i++ {
    defer fmt.Println(i)   // prints 2, 1, 0
}
```

And a `defer` inside a loop does not run at the end of each iteration, it runs when the **function** returns. Opening a thousand files in a loop with `defer f.Close()` keeps all thousand open until the function ends. Put the body in its own function if you need per-iteration cleanup.

PROJECT 3: A WORD COUNTER THAT HANDLES FAILURE

Write a program that takes a filename, counts the words in it, and reports errors properly rather than crashing.

```
package main

import (
    "fmt"
    "os"
    "strings"
)

func countWords(path string) (int, error) {
    data, err := os.ReadFile(path)
    if err != nil {
        return 0, fmt.Errorf("counting words in %s: %w", path, err)
    }
    return len(strings.Fields(string(data))), nil
}

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: wc <file>")
        os.Exit(1)
    }
    n, err := countWords(os.Args[1])
    if err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }
    fmt.Printf("%d words\n", n)
}
```

Run it on a real file, then on one that does not exist, and read the wrapped message. Notice `os.Exit(1)` and writing to `os.Stderr`: a command line tool that reports failure on stdout with exit code zero is a tool that breaks every script that uses it.

`strings.Fields` splits on any run of whitespace, which is the right behaviour here and is worth remembering: it is not the same as `strings.Split(s, " ")`, which gives empty strings between consecutive spaces.

YOUR TURN

Add a check that reports a clearer message when the path is a directory rather than a file. `os.ReadFile` returns an error you can inspect; try `errors.Is(err, syscall.EISDIR)` on Unix, or simply call `os.Stat` first and check `info.IsDir()`, which is portable and is the version you should probably ship.

0

Structs and methods, against objects and classes

A JavaScript object is a bag of keys you can add to at any time. A Go struct is a fixed set of named fields with fixed types, decided when you declare the type and never changing. Everything else in this chapter follows from that one difference.

Declaring and creating

```
type User struct {
    ID    int
    Name  string
    Email string
    Admin bool
}
```

There are four ways to make one, and you will see all four in real code.

```
// 1. Field names. Use this one. It survives a field being added.
u := User{ID: 1, Name: "Ada", Email: "ada@example.com"}

// 2. Positional. Fragile: adding a field breaks every call site.
u := User{1, "Ada", "ada@example.com", false}

// 3. Zero value, then assign.
var u User
u.Name = "Ada"

// 4. new(), giving a pointer to a zero value.
u := new(User)   // *User
```

Notice what is missing from the first example: `Admin` was not set, so it is `false`. There is no `undefined`. Every field always has a value, and a struct you create is immediately complete.

> **IF YOU KNOW JAVASCRIPT**

The closest JavaScript equivalent is a class with a constructor that sets every property. The difference is that you cannot add a field later. `u.Age = 36` on a `User` with no `Age` field is a compile error, not a new property. This is the whole point: the shape of your data is checked, everywhere, for free.

Methods: functions with a receiver

Go has no classes. You attach behaviour to a type by declaring a function with a **receiver**, the parameter in parentheses before the name.

```
func (u User) DisplayName() string {
    if u.Name == "" {
        return u.Email
    }
    return u.Name
}

fmt.Println(u.DisplayName())
```

`u User` is the receiver. It behaves like a parameter, because that is exactly what it is: `u.DisplayName()` is sugar for passing `u` as the first argument.

IF YOU KNOW JAVASCRIPT

There is no `this`, and that removes an entire class of JavaScript bug. The receiver is a normal named parameter, bound at the call site, and it cannot be rebound by how the function was called. No `.bind(this)`, no arrow function workaround, no `const self = this`.

Value receivers and pointer receivers

This is the first thing in Go that genuinely trips people up, and it is worth getting right now rather than debugging later.

A **value receiver** gets a copy. Changes to it do not affect the original.

```
func (u User) Rename(name string) {
    u.Name = name       // modifies the copy, then throws it away
}

u := User{Name: "Ada"}
u.Rename("Grace")
fmt.Println(u.Name)     // still "Ada"
```

A **pointer receiver** gets the address, so changes stick.

```
func (u *User) Rename(name string) {
    u.Name = name       // modifies the original
}

u := User{Name: "Ada"}
u.Rename("Grace")
fmt.Println(u.Name)     // "Grace"
```

You do not write `(&u).Rename(...)`. Go inserts the address-of automatically when the method needs a pointer and you have an addressable value.

> **WATCH OUT**
> The rule in practice: **if a method modifies the receiver, it must be a pointer receiver.** A value receiver that assigns to a field compiles happily and silently does nothing, which is one of the few truly quiet bugs in Go.
>
> The second rule: **be consistent within a type.** If any method on `User` takes a pointer, give them all pointers. Mixing the two is legal and makes the rules about which calls are addressable surface at the worst moment.

A practical reason to prefer pointer receivers even when not mutating: a value receiver copies the whole struct on every call. For a struct with twenty fields called in a loop, that is real work for no benefit.

Composition instead of inheritance

Go has no `extends`. It has **embedding**, which looks similar and behaves differently in one important way.

```
type Timestamps struct {
    CreatedAt time.Time
    UpdatedAt time.Time
}

type Post struct {
    Timestamps          // embedded: no field name
    Title string
    Body  string
}

p := Post{Title: "Hello"}
p.CreatedAt = time.Now()        // promoted, as if declared on Post
fmt.Println(p.Timestamps.CreatedAt)  // the long way, same field
```

The embedded type's fields and methods are **promoted** to the outer type. If `Timestamps` had a method `Age()`, then `p.Age()` would work.

> **IF YOU KNOW JAVASCRIPT**
> This resembles `class Post extends Timestamps`, and the difference matters: there is no polymorphism here. A `Post` is **not** a `Timestamps` and cannot be passed where one is expected. Embedding is a convenience for forwarding fields and methods, not a type relationship. In Go, the thing that lets one type stand in for another is an interface, which is chapter ten.

Struct tags: metadata for encoders

A field can carry a string of metadata, used by libraries through reflection. You will meet this constantly, because it is how Go does JSON.

```
type User struct {
    ID    int    `json:"id"`
    Name  string `json:"name"`
    Email string `json:"email"`
    Admin bool   `json:"-"`                  // never encoded
    Bio   string `json:"bio,omitempty"`      // dropped when empty
}
```

Without tags, `json.Marshal` uses the Go field names, so you get `{"ID":1, "Name":"Ada"}`. With them you get the lowercase JSON your frontend expects.

> **WATCH OUT**
> Those backticks are a raw string literal, and the format inside is strict: `json:"name"` with no space after the colon. `json: "name"` silently does nothing, the tag is ignored, and the field encodes as `Name`. There is no error. If a field is coming out with the wrong key, check for that space first.

Constructors are just functions

Go has no `constructor` keyword. The convention is a package-level function named `New` or `NewThing` that returns the type.

```
func NewUser(name, email string) (*User, error) {
    if email == "" {
        return nil, errors.New("email is required")
    }
    return &User{
```

```
        Name:  name,
        Email: email,
    }, nil
}
```

Returning `*User` rather than `User` is the usual choice, because the caller almost always wants to pass it around without copying, and because it lets you return `nil` on failure.

> **IF YOU KNOW JAVASCRIPT**
> A JavaScript constructor cannot fail cleanly: it either throws or returns a half-built object. A Go constructor is an ordinary function, so it returns an error like everything else, and you cannot get a `*User` back unless it succeeded.

Comparing structs

Structs are comparable with `==` when all their fields are, and the comparison is by value, field by field.

```
a := User{ID: 1, Name: "Ada"}
b := User{ID: 1, Name: "Ada"}
fmt.Println(a == b)     // true
```

That is deep equality for free, and it is a genuine surprise if you are used to `a === b` on objects being an identity check.

> **WATCH OUT**
> It stops working the moment a struct contains a slice, a map or a function, because those are not comparable. The compiler tells you. For those, use `reflect.DeepEqual`, or in tests `google/go-cmp`, which prints a readable diff instead of just `false`.

> **PROJECT 4: A CONTACT BOOK IN MEMORY**
> Build a small type with behaviour, and exercise value and pointer receivers.

```
package main

import (
    "errors"
    "fmt"
    "strings"
)

type Contact struct {
```

```
    Name  string
    Email string
    Tags  []string
}

// Value receiver: reads only, so a copy is fine.
func (c Contact) Display() string {
    if len(c.Tags) == 0 {
        return c.Name
    }
    return fmt.Sprintf("%s [%s]", c.Name, strings.Join(c.Tags, ", "))
}

// Pointer receiver: it modifies, so it must be a pointer.
func (c *Contact) Tag(tag string) {
    for _, t := range c.Tags {
        if t == tag {
            return // already there, do nothing
        }
    }
    c.Tags = append(c.Tags, tag)
}

type Book struct {
    contacts []Contact
}

func (b *Book) Add(c Contact) error {
    if c.Email == "" {
        return errors.New("contact needs an email")
    }
    b.contacts = append(b.contacts, c)
    return nil
}

func (b *Book) Find(email string) (*Contact, bool) {
    for i := range b.contacts {
        if b.contacts[i].Email == email {
            return &b.contacts[i], true   // address of the real one
        }
    }
    return nil, false
}
```

```
func main() {
    var book Book
    if err := book.Add(Contact{Name: "Ada", Email:
"ada@example.com"}); err != nil {
        fmt.Println(err)
        return
    }

    if c, ok := book.Find("ada@example.com"); ok {
        c.Tag("maths")
        c.Tag("maths") // ignored, already present
    }

    for _, c := range book.contacts {
        fmt.Println(c.Display())  // Ada [maths]
    }
}
```

The subtle line is `return &b.contacts[i], true`. Ranging with `for _, c := range b.contacts` would give you a **copy** of each contact, and tagging that copy would change nothing. Indexing with `i` and taking the address reaches the real element. Change it to the copying version and watch the tag vanish: that is the lesson of this chapter in one experiment.

YOUR TURN

Add a `Remove(email string) bool` method to `Book`. You will need to build a new slice or use the `slices.Delete` helper from the standard library. Then add a `Count()` method and decide, with a reason you can say out loud, whether it takes a value or a pointer receiver.

0

Slices: the array that is not an array

Go has arrays and it has slices. You will use slices for essentially everything, and the distinction matters exactly twice: when you declare one, and when a surprising bug appears. This chapter covers both.

Arrays have a fixed length, in the type

```
var a [3]int        // an array of exactly three ints
a[0] = 1
fmt.Println(len(a)) // 3
```

The length is **part of the type**. `[3]int` and `[4]int` are different types, and a function taking `[3]int` will not accept `[4]int`. Arrays are also copied whole on assignment. This is almost never what you want, which is why arrays are rare in Go code.

Slices are a view onto an array

```
nums := []int{1, 2, 3}      // no length in the brackets: a slice
nums = append(nums, 4)
fmt.Println(len(nums))      // 4
```

A slice is a small three-field value: a pointer to an underlying array, a length, and a capacity. That is the whole model, and understanding it explains every slice surprise you will ever hit.

`len(s)`	How many elements you can index right now
`cap(s)`	How many fit before the underlying array must be reallocated

```
s := make([]int, 2, 5)   // len 2, cap 5
fmt.Println(len(s), cap(s))  // 2 5
s = append(s, 10)            // fits in the existing array
fmt.Println(len(s), cap(s))  // 3 5
```

`make([]T, len, cap)` creates one with room reserved. When you know roughly how many elements are coming, say so: it avoids repeated reallocation.

> **IF YOU KNOW JAVASCRIPT**
> A JavaScript array handles all of this invisibly. `push` grows it and you never think about capacity. Go exposes the machinery, which costs you some attention and buys you control over allocation, which matters a great deal in a hot loop and not at all in a handler that runs once per request.

append, and the thing it does not promise

`append` returns a slice. **You must use the return value.**

```
nums := []int{1, 2, 3}
append(nums, 4)          // compile error: result not used
nums = append(nums, 4)   // correct
```

The reason is that `append` may or may not allocate a new array. If there is spare capacity it writes in place and returns a slice with a longer length. If there is not, it allocates a bigger array, copies everything over, and returns a slice pointing at the **new** array. The caller's original slice still points at the old one.

This is where the famous slice bug lives.

```
a := []int{1, 2, 3, 4, 5}
b := a[1:3]              // [2 3], sharing a's array
b[0] = 99
fmt.Println(a)           // [1 99 3 4 5]  <- a changed
```

Slicing does not copy. `b` is a window onto the same memory. Writing through one is visible through the other.

> **WATCH OUT**
> It gets worse with `append`, because whether you corrupt the original depends on capacity, which depends on history:
>
> ```
> a := []int{1, 2, 3, 4, 5}
> b := a[1:3]              // len 2, cap 4 (to the end of a)
> b = append(b, 99)        // fits in cap, writes into a's array
> fmt.Println(a)           // [1 2 3 99 5]  <- clobbered a[3]
> ```
>
> The fix, when you want a genuinely independent slice, is a three-index slice that caps capacity, or an explicit copy:

```
b := a[1:3:3]            // cap is now 2, so append must allocate
// or
b := slices.Clone(a[1:3])
```

This bug is rare in everyday handler code and vicious when it appears. The tell is "a value changed and nothing in my code assigned to it".

Iterating

```
for i, v := range nums {
    fmt.Println(i, v)
}

for _, v := range nums {    // index discarded
    fmt.Println(v)
}

for i := range nums {       // value discarded
    nums[i] *= 2
}
```

`range` gives you a **copy** of each element. For ints that is irrelevant. For structs it is the bug from the previous chapter: modifying `v` changes nothing. Use the index when you need to write.

IF YOU KNOW JAVASCRIPT

Go has no `map`, `filter`, `reduce`, `find` or `some` for slices in the language. Since Go 1.21 the `slices` package covers some of it:

```
slices.Contains(nums, 3)        // bool
slices.Index(nums, 3)           // position or -1
slices.Sort(nums)               // in place
slices.Reverse(nums)            // in place
slices.Max(nums)                // and Min
```

For anything else, write the loop. A filter is four lines:

```
var active []User
for _, u := range users {
    if u.Active {
        active = append(active, u)
```

```
    }
}
```

Passing slices to functions

A slice is cheap to pass: you are copying three words, not the data. That means a function **can** modify the caller's elements.

```
func zero(nums []int) {
    for i := range nums {
        nums[i] = 0
    }
}

nums := []int{1, 2, 3}
zero(nums)
fmt.Println(nums)   // [0 0 0]
```

But it cannot change the caller's **length**, because the slice header itself was copied:

```
func addOne(nums []int) {
    nums = append(nums, 1)   // changes the local copy's header only
}

nums := []int{1, 2, 3}
addOne(nums)
fmt.Println(len(nums))   // still 3
```

The convention, and the reason `append` works the way it does: a function that grows a slice returns it.

```
func addOne(nums []int) []int {
    return append(nums, 1)
}
nums = addOne(nums)
```

Sorting

```
people := []User{{Name: "Zoe"}, {Name: "Ada"}}

slices.SortFunc(people, func(a, b User) int {
```

```
    return strings.Compare(a.Name, b.Name)
})
```

The comparison function returns a negative number, zero, or a positive number, the same contract as JavaScript's `Array.prototype.sort` comparator. For stable sorting use `slices.SortStableFunc`.

PROJECT 5: WORD FREQUENCY, TOP TEN

Read a file, count how often each word appears, and print the ten most common. This uses slices, sorting and a map together, and it is a real tool you can point at a book.

```
package main

import (
    "fmt"
    "os"
    "slices"
    "strings"
)

type pair struct {
    word  string
    count int
}

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: freq <file>")
        os.Exit(1)
    }

    data, err := os.ReadFile(os.Args[1])
    if err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }

    counts := map[string]int{}
    for _, w := range strings.Fields(string(data)) {
        w = strings.ToLower(strings.Trim(w, ".,!?;:\"'()"))
        if w != "" {
            counts[w]++
        }
    }
```

```
    pairs := make([]pair, 0, len(counts))
    for w, c := range counts {
        pairs = append(pairs, pair{w, c})
    }

    slices.SortFunc(pairs, func(a, b pair) int {
        if b.count != a.count {
            return b.count - a.count       // count descending
        }
        return strings.Compare(a.word, b.word) // then alphabetical
    })

    for _, p := range pairs[:min(10, len(pairs))] {
        fmt.Printf("%6d  %s\n", p.count, p.word)
    }
}
```

Three things worth noticing. `make([]pair, 0, len(counts))` reserves capacity up front because we know exactly how many there will be. The tie-break on word makes the output deterministic, which matters because map iteration order is random and without it two runs would disagree. And `min` is a builtin since Go 1.21, so `pairs[:min(10, len(pairs))]` does not panic on a file with fewer than ten distinct words.

YOUR TURN

Add a flag so the user can ask for a different number of results, for example `freq book.txt 25`. Then add a stop-word list so "the", "a" and "and" are excluded, and decide whether a slice or a map is the better structure for holding those stop words. The next chapter will tell you whether you chose right.

0

Maps, and why iteration order is random on purpose

A Go map is a hash table with typed keys and typed values. It is the closest thing Go has to a JavaScript object used as a dictionary, and it differs in four ways that each cause a specific bug the first time you meet them.

Declaring and using

```
ages := map[string]int{
    "ada":   36,
    "grace": 45,
}

ages["alan"] = 41
fmt.Println(ages["ada"])    // 36
fmt.Println(len(ages))      // 3
delete(ages, "alan")
```

The type is `map[KeyType]ValueType`. Keys must be comparable, so strings, ints, booleans and structs of those are fine; slices, maps and functions are not.

WATCH OUT

A `nil` map is readable but **not writable**, and this is the first map bug everyone hits:

```
var m map[string]int      // nil
fmt.Println(m["a"])       // 0, fine
fmt.Println(len(m))       // 0, fine
m["a"] = 1                // panic: assignment to entry in nil map
```

Always create one before writing: `m := map[string]int{}` or `m := make(map[string]int)`. The same trap hides inside structs, where a map field is nil until you assign one, so a constructor that forgets it produces a struct that panics on first write.

Missing keys return the zero value

Reading a key that is not there is not an error. You get the zero value of the value type.

```
counts := map[string]int{}
fmt.Println(counts["nothing"])   // 0, not undefined
counts["word"]++                 // works: reads 0, writes 1
```

That last line is why counting in Go is so tidy: no “if the key exists, add one, otherwise set it to one”.

To tell “absent” from “present but zero”, use the two-value form:

```
v, ok := ages["nobody"]
if !ok {
    fmt.Println("not in the map")
}
```

`ok` is a bool. This is the **comma ok** idiom, and you will see the same shape for type assertions and channel receives.

> **IF YOU KNOW JAVASCRIPT**
> In JavaScript, `obj.missing` gives `undefined`, and you distinguish absent from set-to-undefined with `in` or `hasOwnProperty`. Go’s `v, ok :=` is the same idea with better ergonomics, because the check and the read are one operation and you cannot forget which you did.

Iteration order is deliberately random

```
for k, v := range ages {
    fmt.Println(k, v)
}
```

Run that twice and the order differs. This is not an implementation accident that might settle down: the runtime randomises the starting point on purpose, so that code which accidentally depends on order fails immediately and loudly rather than in production two years later when the table grows.

To iterate in a stable order, sort the keys:

```
keys := make([]string, 0, len(ages))
for k := range ages {
    keys = append(keys, k)
}
slices.Sort(keys)
```

```
for _, k := range keys {
    fmt.Println(k, ages[k])
}
```

> **IF YOU KNOW JAVASCRIPT**
> JavaScript objects do have an order: integer-like keys ascending, then string keys in insertion order. Plenty of code quietly relies on it. Go removes the temptation. If your output needs an order, you must say what that order is, which is usually an improvement to the code anyway.

Maps are reference types

Passing a map to a function does not copy it. The function can modify the caller's map.

```
func add(m map[string]int) {
    m["new"] = 1        // visible to the caller
}
```

This is unlike a struct and unlike an array, and like a slice's elements. There is no `const` map and no way to pass one read-only. If a function must not modify a map, the only guards are a copy (`maps.Clone`) or a comment.

Structs as values, and the thing you cannot do

A map of structs has one sharp edge:

```
type Stat struct{ Count int }
m := map[string]Stat{"a": {Count: 1}}

m["a"].Count++      // compile error: cannot assign to struct field
```

The map returns a **copy** of the struct, so incrementing it would modify something about to be discarded, and Go refuses rather than letting you write code that does nothing. Two fixes:

```
// 1. Read, modify, write back.
s := m["a"]
s.Count++
m["a"] = s

// 2. Store pointers instead.
```

```
m := map[string]*Stat{"a": {Count: 1}}
m["a"].Count++      // fine
```

The second is usually what you want when the values are large or frequently updated. The first is fine for small structs and avoids a layer of indirection.

Sets

Go has no set type. The idiom is a map with an empty struct value:

```
seen := map[string]struct{}{}
seen["ada"] = struct{}{}

if _, ok := seen["ada"]; ok {
    fmt.Println("already seen")
}
```

`struct{}` is a struct with no fields and it occupies zero bytes, so the map stores only keys. Using `map[string]bool` is also common and slightly more readable at the cost of one byte per entry; both are accepted Go.

> **IF YOU KNOW JAVASCRIPT**
> JavaScript has a real `Set` with `.has()` and `.add()`. Go's version is wordier, and that `struct{}{}` reads as line noise until you have seen it a dozen times. Read it as: the type `struct{}`, then an empty literal of it.

Maps are not safe for concurrent use

Two goroutines writing to the same map at the same time will crash the program with a runtime error, deliberately. Go detects it and stops rather than corrupting the table silently.

The answer is usually a mutex, which is chapter fifteen:

```
type Counter struct {
    mu     sync.Mutex
    counts map[string]int
}

func (c *Counter) Inc(key string) {
    c.mu.Lock()
    defer c.mu.Unlock()
    c.counts[key]++
}
```

IF YOU KNOW JAVASCRIPT
This has no JavaScript equivalent, because the event loop means two pieces of your code never run at literally the same instant. In Go they do. Any value shared between goroutines needs protection, and a map is the one where forgetting is immediately fatal rather than subtly wrong, which is mercifully the better failure.

PROJECT 6: A WORD INDEX WITH LINE NUMBERS
Extend the frequency counter into something genuinely useful: for each word, record every line it appears on. This is a `map[string][]int`, a map whose values are slices, and it is a shape you will use often.

```
package main

import (
    "bufio"
    "fmt"
    "os"
    "slices"
    "strings"
)

func main() {
    if len(os.Args) < 2 {
        fmt.Fprintln(os.Stderr, "usage: index <file>")
        os.Exit(1)
    }

    f, err := os.Open(os.Args[1])
    if err != nil {
        fmt.Fprintln(os.Stderr, "error:", err)
        os.Exit(1)
    }
    defer f.Close()

    index := map[string][]int{}

    scanner := bufio.NewScanner(f)
    line := 0
    for scanner.Scan() {
        line++
        for _, w := range strings.Fields(scanner.Text()) {
            w = strings.ToLower(strings.Trim(w, ".,!?;:\"'()"))
            if len(w) < 4 {
```

```
                continue    // skip short words
            }
            // Appending to a nil slice works, so no "if absent"
branch.
            if !slices.Contains(index[w], line) {
                index[w] = append(index[w], line)
            }
        }
    }
    if err := scanner.Err(); err != nil {
        fmt.Fprintln(os.Stderr, "reading:", err)
        os.Exit(1)
    }

    words := make([]string, 0, len(index))
    for w := range index {
        words = append(words, w)
    }
    slices.Sort(words)

    for _, w := range words {
        if len(index[w]) > 2 {   // only words on three or more lines
            fmt.Printf("%-16s %v\n", w, index[w])
        }
    }
}
```

The line that rewards attention is `index[w] = append(index[w], line)`. If `w` is not in the map, `index[w]` is a nil slice, and appending to a nil slice is legal and returns a one-element slice. So the "create the entry if missing" branch that you would write in most languages simply does not exist here. Two zero-value rules, slices and maps, combining to delete a line of code.

Also note `bufio.Scanner`: it reads line by line without loading the file into memory, so this works on a file larger than your RAM, unlike the `os.ReadFile` in the last project. And `scanner.Err()` must be checked, because the loop ends identically on end-of-file and on a read error.

YOUR TURN

Change the value type to a small struct holding both the line numbers and a total count, and watch the compiler stop you from writing `index[w].Count++`. Fix it both ways described above, then decide which you prefer and why. That decision, made consciously once, is the difference between copying Go idioms and understanding them.